

Physik II — FS 2024 — Prof. R. Wallny — Serie 7

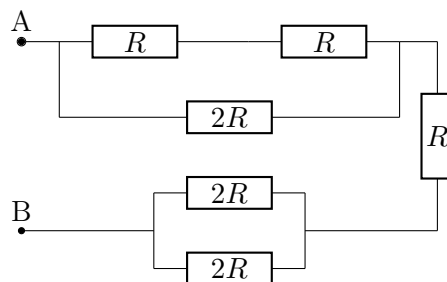
Release: 9th April 2024 – Due date and discussion: 18th April 2024

Exercise 1: Networks of resistors

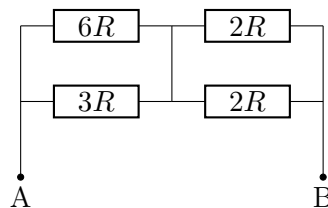
Goal: Understand networks of resistors.

Calculate the equivalent resistance between the points A and B for the shown networks of resistors.

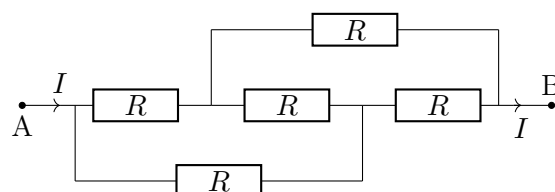
a) Combination 1:



b) Combination 2:



c) Combination 3:



Hint: Assume a current I going through the whole network. It could be helpful to figure out how much current goes through each resistor individually.

Exercise 2: Practical examples

Goal: Difference between ohmic resistance and light bulb. Internal resistance.

An old light bulb for $V = 230\text{ V}$ uses a coiled tungsten wire with a diameter of $d = 40\text{ }\mu\text{m}$ and a length of $l = 1.5\text{ m}$.

- What are the values for the resistance at room- vs. operating temperature of 2700°C ? Use the specific resistance of tungsten $\rho = 0.055 \times 10^{-6}\text{ }\Omega\text{ m}$ at room temperature and its linear temperature dependence of $\alpha = 4.5 \times 10^{-3}\text{ K}^{-1}$. Assume that the dimensions of the wire stay constant.
- Which current goes through the wire immediately after turning it on and during normal operation. After approximately 1000 hour these light bulbs tend to break. In which process does the breaking occur the most?
- How much power in the form of light is emitted if the efficiency is around $\eta = 5\%$? Where does the rest of the power go?

A car battery of a gasoline powered car in normal operation has a voltage of $V_0 = 12\text{ V}$. When the engine is started this voltage drops to a value of $V_1 = 10\text{ V}$ and a current of $I = 150\text{ A}$ is flowing.

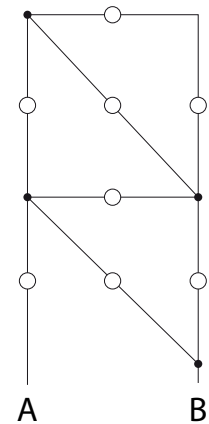
- How big are the resistance of the starter R_a and the internal resistance of the battery R_i ? Draw a matching equivalent circuit diagram.
- In winter, at low temperatures the internal resistance rises and becomes equal the starter resistance $R_a = R_i$. What is the value of voltage V_1 in this case?
- Calculate the power dissipated from the battery and starter in the two cases of the two previous exercises.

Exercise 3: Switching circuit with capacitors

Objective: Calculate networks of capacitors vs resistances.

Points A and B are the endpoints of a network of eight elements.

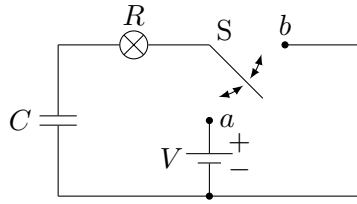
- How large is the total capacitance if the elements are capacitors of equal capacitance C ?
- How large is the total resistance if the elements are resistors of equal resistance R ?



Exercise 4: Circuit

Objective: Kirchhoff's rules, solving differential equations.

Consider the circuit in the figure below. An ideal battery with constant voltage V is connected to a capacitor C and a lamp of resistance R . In the beginning the capacitor is not charged. The switch S is first connected to contact a and left there for a long time (long enough to reach stationary state). At time $t = 0$ the switch S is moved to the contact b .



- Qualitatively describe what happens with the lamp when the switch is first connected with contact a and later switched to contact b .
- Calculate the current I through the lamp of resistance R as a function of t for $t > 0$.
Hint: Write down Kirchhoff's loop rule and derive the differential equations from there.
- Calculate the total energy that is dissipated in the lamp between $t = 0$ and the much later time t_s . How big is this energy compared to the energy stored in part a)?

Exercise 5: Galilean transformations

Objective: Galilean-transformation

Two firecrackers F1 and F2 (see figure) are separated by a distance d from each other and form an angle α with a rail road running between them. Both firecrackers explode while a train moves past at a velocity v .

- Assume that firecracker F1 explodes Δt before the F2. What is the (spatial) distance between the two explosions measured in the train's reference frame?
Hint: Choose the point between the two firecrackers as the origin of the inertial frame.
- For what value of α will this distance be minimal? For what value will it be maximal? Discuss the results.

